

Precise Chrome Etching in Downstream Chlorine Plasmas with Electron Depletion Through Negative Ion Production

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In micropatterning down to 0.1 μm class devices, photolithography requires highly accurate microfabrication of chrome (Cr) reticle masks. In this paper we report the vertical chrome etching using electron beam (EB) resist masks in downstream chlorine plasmas, where the resist damage during etching can be suppressed because of the reduced electron density through negative ion production. Though the Cr etch rate in pure Cl_2 downstream plasmas was only 15 $\text{\AA}/\text{min}$, slight oxygen addition of 1% resulted in a great increase of Cr etch rates as large as 20 times, while resist etch rates increased only twice. Though isotropic Cr etching reactions with chlorine and oxygen radicals caused undercut features, appropriate substrate biasing allows vertical patterning of 0.2 μm wide lines and spaces due to sidewall protection by redeposition of by-products from etched Cr and resists.

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In micropatterning down to 0.1 μm class large scale integrated (LSI) devices, photolithography also requires highly accurate microfabrication of four or five times magnification of chrome (Cr) reticle masks. Optical proximity effect correction (OPC), which is a promising methodology to improve critical dimension (CD) control and increase overlay margins for the shorter exposure wavelength, requires the fabrication of photomasks with 0.1–0.2 μm size patterns.¹ Since conventionally used wet etching suffers from essential resolution limit due to its isotropic etch reaction, dry etching with smaller CD loss and higher fidelity is desired for the next-generation Cr mask making.^{2,3} So far, however, the low dry etching resistance of electron beam (EB) resists has prevented the practical use of dry etching. On the other hand, in the last several years, we have studied novel plasma processes using negative ions generated in halogen downstream plasmas, *e.g.*, charging-free gate poly-Si etching in SF_6 and high-rate aluminum etching in Cl_2 .^{4,5} In this study we have investigated the chrome etching using EB resist masks in downstream chlorine plasmas, where electron irradiation damage can be suppressed owing to the reduced electron density resulting from negative ion production.

Experimental

Etched samples were Cr films on Si wafers instead of real quartz mask blanks due to the feasibility of cleaving for cross-sectional scanning electron microscopy (SEM) observation of etched features. Cr films of approximately 1000 $\text{\AA}$ thickness were sputter-deposited on 7000 $\text{\AA}$ oxide thermally grown on Si wafers. Line and space (L&S) patterns with 0.6 μm height and 0.2 μm width in negative-type EB resist (Shipley SAL®601) were defined using 75 keV EB delineator (Elionix ELS-7700) on Cr/Si with oxide films. As shown in Fig. 1, chlorine inductively coupled plasmas were produced in a stainless steel chamber of 260 mm in diameter by feeding a 13.56 MHz rf power to a one-turn antenna set on a quartz window at one end and a water-cooled stage for etching was set in the downstream region 14–20 cm from the antenna. The stage was biased with capacitively coupled 400 kHz HF. Samples were fixed on the stage with a wafer-to-stage gap of approximately 100 μm by inserting a polyimide film spacer in the peripheral region of the wafer since, in the actual reticle making, it cannot be chucked directly on the sample electrode and might require helium back side cooling to control the temperature during etching. Prior to etching experiments the peak-to-peak bias voltage on the sample surface in Cl_2 downstream plasmas (V_{p-p}) was directly measured using an oscilloscope as a function of voltages at the stage electrode (V_0). Though the V_0 value was actually monitored in etching experiments, data in the following sections are presented with respect to V_{p-p} .

Results and Discussion

Production of negative ion-rich chlorine plasmas.—At first, conditions for generating negative ion-rich Cl_2 plasmas were investigated using an electrostatic probe. As shown in Fig. 2a and b, the probe saturation current ratio I_+/I_- was measured as a function of the distance from the antenna (L) and the pressure, respectively. I_+ and I_- are saturation currents at probe potentials of +40 V and –40 V, respectively. Namely, I_+ consists of both electrons and negative ions, while I_- is substantially positive ion current. The decrease of I_+/I_- value observed at longer distances L and higher pressures implies the decrease of electron density ratio in plasmas through negative ion production.^{4,5} In general, at higher pressures or longer distances L , namely, in downstream plasmas, electron temperatures become lower through collisions in the gas phase. Then, electron dissociative attachment Reaction 1 by low-energy electrons dominantly occurs and electron impact detachment Reaction 2 by high-energy electrons is reduced, consequently resulting in the increase of negative ions

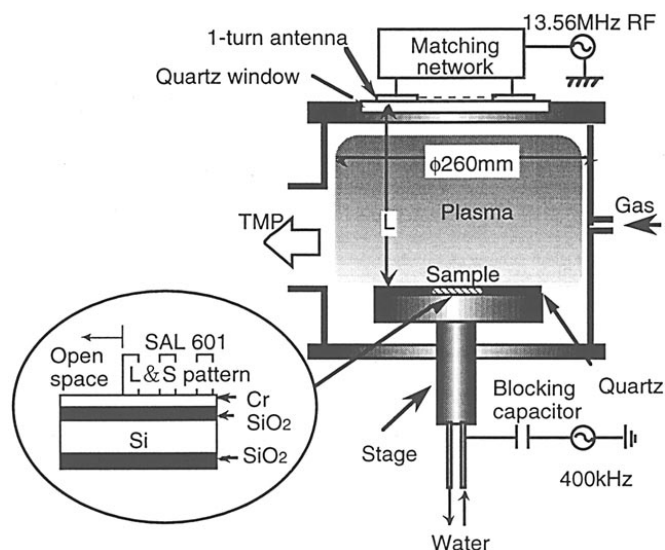
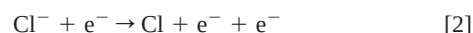
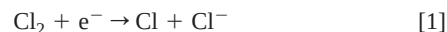


Figure 1. A planar-type inductively coupled plasma etcher employed in this study.

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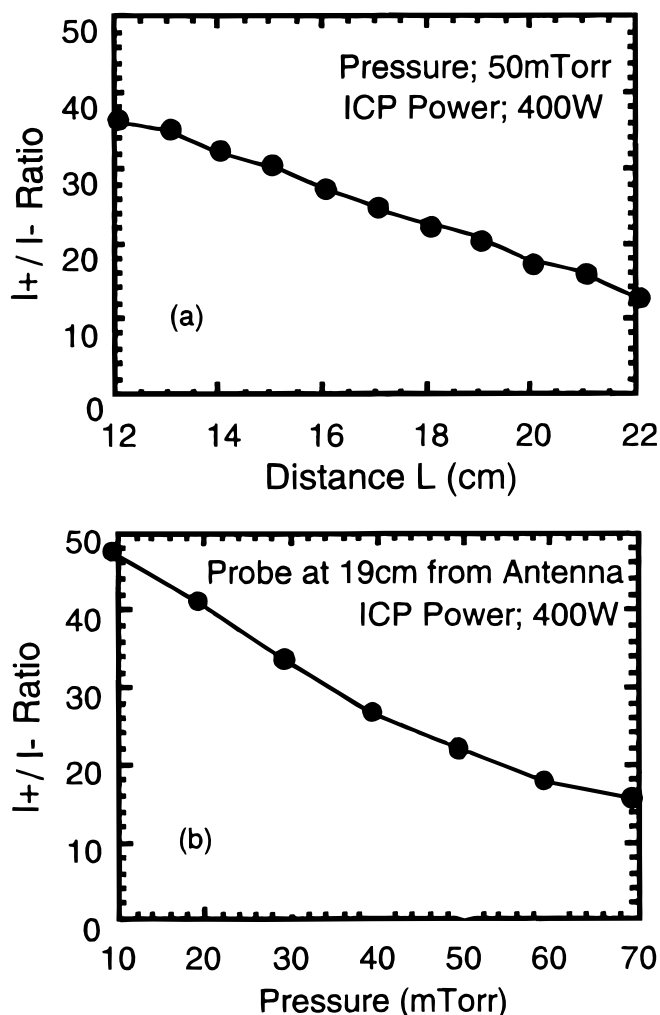


Figure 2. Saturation current ratio I_+/I_- as a function of the distance from the antenna L (a, top) and the pressure (b, bottom), where I_+ and I_- are saturation currents when an electrostatic probe was biased at +40 V and -40 V, respectively.

In our etching experiments we used the following conditions for producing negative ion-rich plasma unless otherwise noted; 400 W ICP power, 50 mTorr pressure, and 19 cm distance L , where electron density is less than several percentages relative to process plasmas commonly used.

Etching characteristics in downstream chlorine plasmas.—Figure 3 shows the substrate position dependence of Cr and EB resist etch rates in pure Cl_2 plasmas at 50 mTorr. Substrate biasing was kept constant so as to induce the peak-to-peak bias voltage on the substrate surface V_{p-p} of 45 V for each data set. Resist etch rates significantly decreased with the increase of the distance L , while those of Cr are scarcely influenced by the distance except for the datum at $L = 14$ cm, where high flux irradiation of electrons and ions might cause a serious temperature rise. These etching characteristics can be explained in connection with the decrease of plasma density with the distance L . Since the resist etching is dominated by sputtering, its etch rate decreases with the decrease of plasma density. On the contrary, the Cr etch rate is limited by surface reactions with chlorine radicals, and hence it does not depend on the plasma density.

Figure 4 shows the EB resist shrinkage width after etching as a function of the distance L . Here, the resist shrinkage is defined as a difference between the original resist width L_1 and the resist width after plasma irradiation L_2 , as inset in the figure. Notable shrinkage of EB resist was observed for the upstream case, while it was suppressed for the downstream case, resulting in a smaller CD shift.

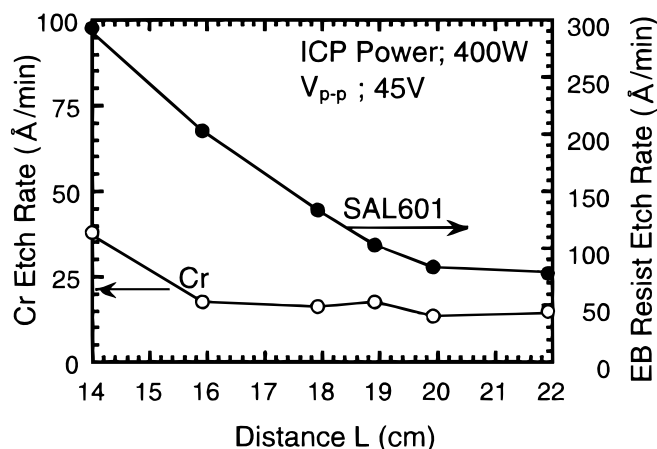


Figure 3. Substrate position dependence of Cr and EB resist etch rates in Cl_2 plasmas at 50 mTorr.

Thus the deformation of EB resists can be suppressed in the downstream plasmas due to the reduced electron irradiation flux through negative ion production.

Moreover, Fig. 5 shows the pressure dependence of Cr and EB resist etch rates in Cl_2 downstream plasmas. Higher resist etch rates were observed at lower pressures, namely, under conditions for the higher electron density. On the other hand, Cr etch rates were not so much influenced by the plasma density variation with pressures. The low Cr etch rate at 10 mTorr is possibly due to the shortage of Cl radical flux. Such etching characteristics are consistent with the above explanation for Fig. 3.

Next, Fig. 6 shows the substrate bias dependence of Cr and EB resist etch rates in Cl_2 downstream plasmas. Linear increase of resist etch rates with the substrate bias voltage V_{p-p} indicates that the resist etch rate is dominated by the ion-bombardment-enhanced desorption reaction. On the other hand, Cr etching did not occur without supplying any bias voltages. Thus reaction products between Cr and Cl cannot desorb readily from surfaces without ion bombardment as expected from the high melting point of CrCl_2 at 830°C. When the bias voltage V_{p-p} was higher than 10 V, Cr etch rates were almost constant at 15 $\text{\AA}/\text{min}$ irrespective of bias voltages, indicating that surface reactions with radicals rather than ion-bombardment-enhanced reactions are rate limiting. As for the etch rate variation with the pattern density, etch rates at L&S patterns were always

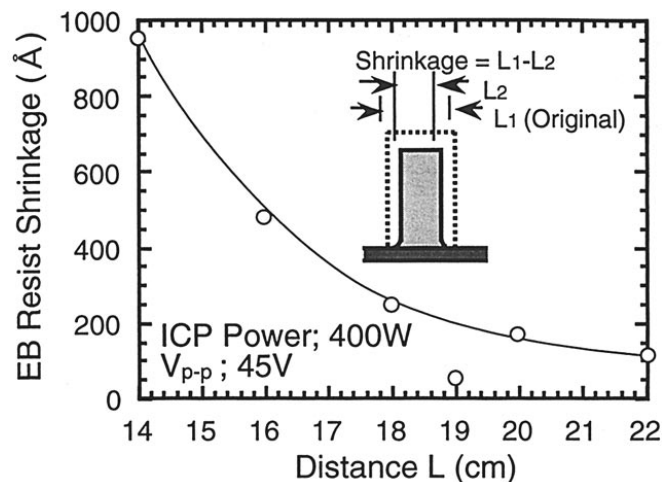


Figure 4. Substrate position dependence of the EB resist shrinkage after etching. Resist shrinkage is defined as the difference between the original resist width L_1 and the resist width after plasma irradiation L_2 , as inset in the figure.

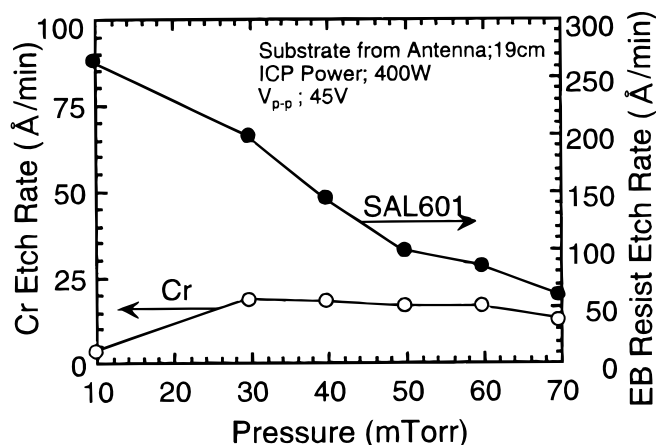


Figure 5. Pressure dependence of Cr and EB resist etch rates in Cl_2 downstream plasmas.

lower than those at open spaces owing to reduced ion and radical flux in high aspect ratio resist patterns and redeposition of resist sputtered materials.

Oxygen addition effects on Cr etching characteristics.—With the purpose of increasing the Cr etch rate, oxygen addition effects on Cr and resist etch rates were investigated as shown in Fig. 7. It is reported that oxygen and the Cl radical react with Cr to form CrO_2Cl_2 (boiling point, 117°C) which is possibly volatile when a substrate temperature is raised by the plasma exposure.⁶ In high density plasmas used in this study a slight oxygen addition of 5×10^{-4} Torr, i.e., 1% of the total pressure, resulted in the great increase of the Cr etch rate as large as 20 times. It was anticipated that the oxygen addition may drastically enhance resist etching reactions. Indeed resist etch rates increased linearly up to 1×10^{-4} Torr, i.e., 0.2%, where reaching twice the value for the no-addition case, but then saturated for further addition. This saturation of resist etch rates is explained as follows. Oxygen radicals react with the resist surface to form reaction products which can desorb easily under ion bombardment, causing the linear increase of resist etch rates up to 0.2% oxygen addition. At the oxygen addition of 0.2%, however, reaction sites on the resist surface are fully occupied. Hence resist etch rates are limited by the ion-enhanced desorption at an oxygen addition above 0.2%, resulting in the saturated resist etch rates. Though someone might guess this saturated resist etch rate is due to the redeposition of etching by-products on the resist, this possibility is denied as discussed

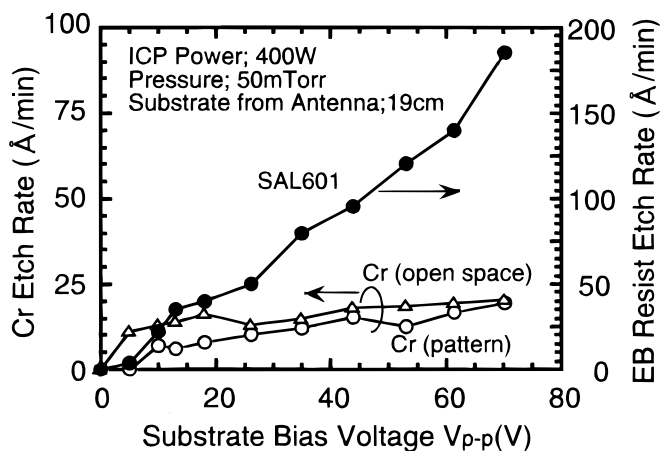


Figure 6. Substrate bias dependence of Cr and EB resist etch rates in Cl_2 downstream plasmas.

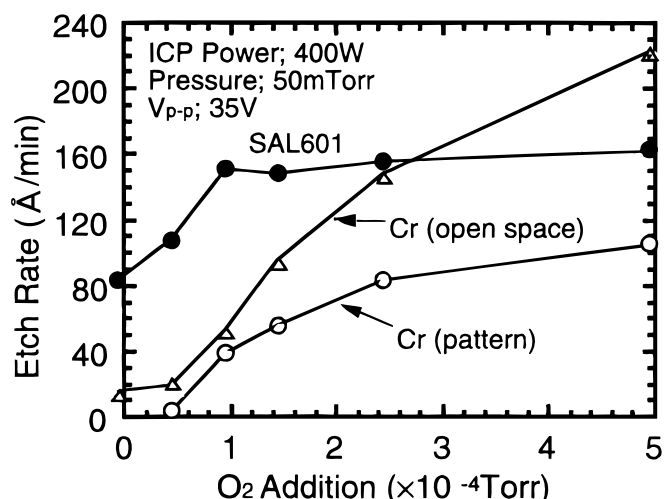


Figure 7. Cr and EB resist etch rates as a function of oxygen addition in Cl_2 plasmas. Etching conditions are 19 cm distance from the antenna, 400 W ICP power, 50 mTorr pressure, and 35 V peak-to-peak substrate bias.

in the last part of this section. Cr etch rates at L&S patterns that are lower than those at open spaces are ascribed to the reduced radical flux and redeposition of the sputtered resist.

Substrate bias dependence of Cr and EB resist etch rates was also investigated in $\text{Cl}_2/0.5\%$ O_2 downstream plasmas as shown in Fig. 8. At open spaces Cr etching occurred even at the floating potential. Cr etch rates at both L&S patterns and at open spaces increased with V_{p-p} up to 20 V and then saturated for further increase of V_{p-p} . This means that ion-enhanced desorption of the reaction products is the rate-limiting process below 20 V, while above 20 V are surface reactions with radicals similar to the results of Fig. 6.

Since spontaneous etching occurs when oxygen is added, undercutting may cause the linewidth loss. Figure 9a shows substrate bias dependence of the Cr undercut width and the EB resist etched depth after etching in $\text{Cl}_2/0.5\%$ O_2 plasmas for 20 min. Typical cross-sectional SEM photographs of lines and spaces etched in $\text{Cl}_2/0.5\%$ O_2 plasmas for 20 min at different biasing voltages are also shown in Fig. 9b. Estimates using the Cr etch rate of 80 Å/min at the patterned area, etching time of 20 min corresponds to the overetch time of approximately 45% at L&S patterns and approximately 200% in the open spaces, where etch rates are around twice as high as those in the patterned area. Since the density of radicals that attacks the sidewall

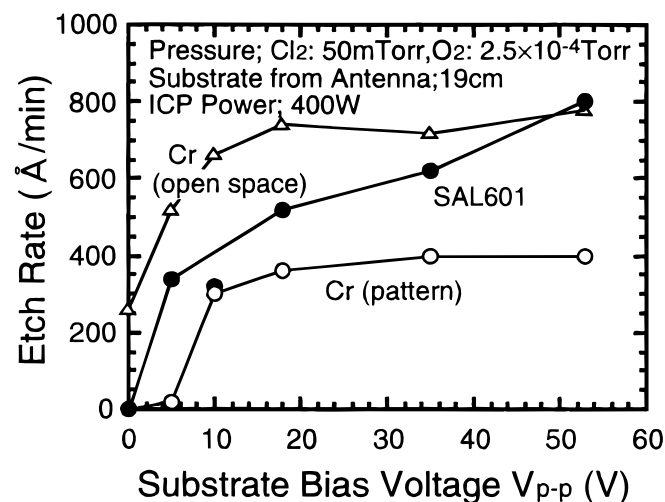
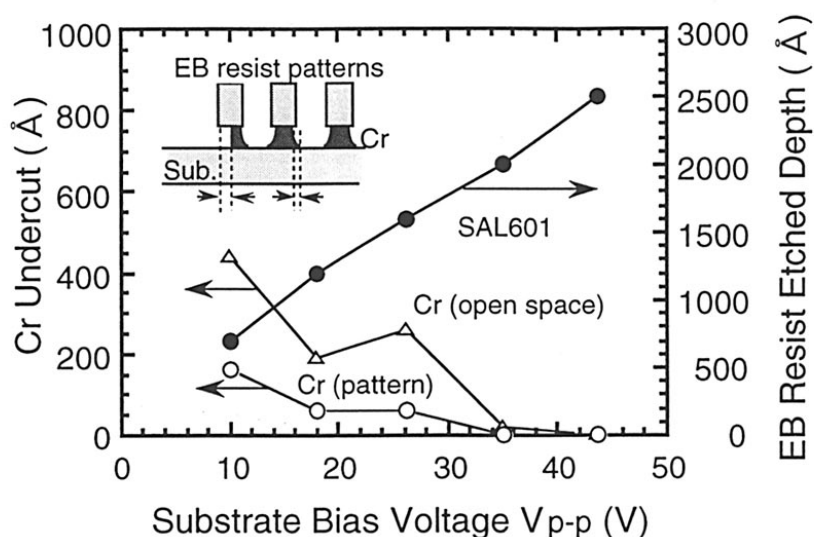


Figure 8. Substrate bias dependence of Cr and EB resist etch rates in $\text{Cl}_2/0.5\%$ O_2 downstream plasmas.

(a)



(b)

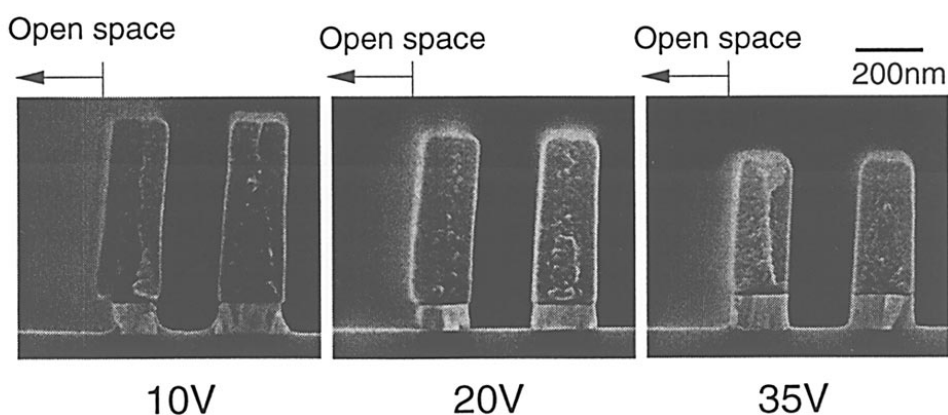


Figure 9. (a) The Cr undercut length and the EB resist etch depth after etching in $Cl_2/0.5\% O_2$ plasmas for 20 min as a function of the peak-to-peak substrate bias voltage V_{p-p} . Etching conditions were ICP power; 400 W; pressure, 50 mTorr; and distance L , 19 cm. (b) SEM photographs of etched patterns at $V_{p-p} = 10$, 20, and 35 V.

increases during the overetch period, notable undercut was observed especially at lines beside open spaces. Undercut features were apparently observed at low bias voltages, and were suppressed when bias

voltage was increased to form the sidewall protective deposition from sputtered resist materials and Cr etch products. Finally the vertical etched features were obtained at $V_{p-p} = 35$ V.

Furthermore, time evolution of etched features at $V_{p-p} = 35$ V was examined as shown in Fig. 10. Cr films with 1100 Å thickness were completely etched within 15 min. The linear increase of the resist etched depth indicates that resist etch rates are not affected by the re-deposition of etch products. Figure 11 shows SEM photographs of L&S patterns etched for 20 and 40 min. Excess overetch caused undercut features propagated from the bottom end of the sidewall, where the sidewall deposition is thinnest.

Conclusions

Highly accurate Cr dry etching in chlorine downstream plasmas is summarized as follows. Reaction products between Cr and Cl radicals do not desorb readily without biasing substrate, and even under energetic ion bombardment etch rates are still low at 15 Å/min due to the rate-limiting surface reactions with Cl radicals. The slight addition of oxygen allows the drastic increase in the Cr etch rate due to the spontaneous etching reactions with Cl and O radicals, and consequently 20 times higher Cr etch rate was achieved by oxygen addition of only 1%. Vertical etched features can be obtained when the sidewall protective deposition is formed from etching by-products. On the other hand, in general, EB resists show poor dry etching resis-

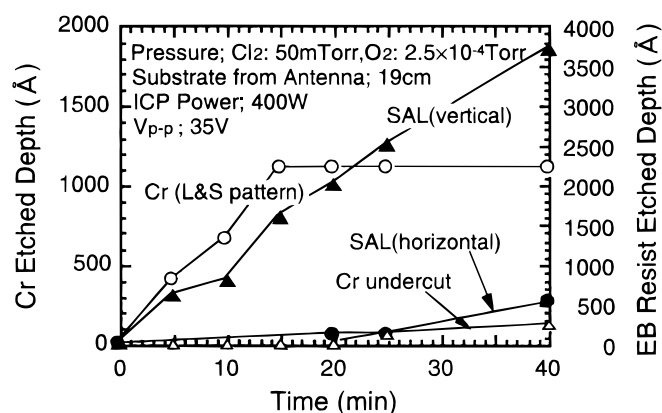


Figure 10. Time evolution of L&S pattern etched features at $V_{p-p} = 35$ V in $Cl_2/0.5\% O_2$ downstream plasmas.

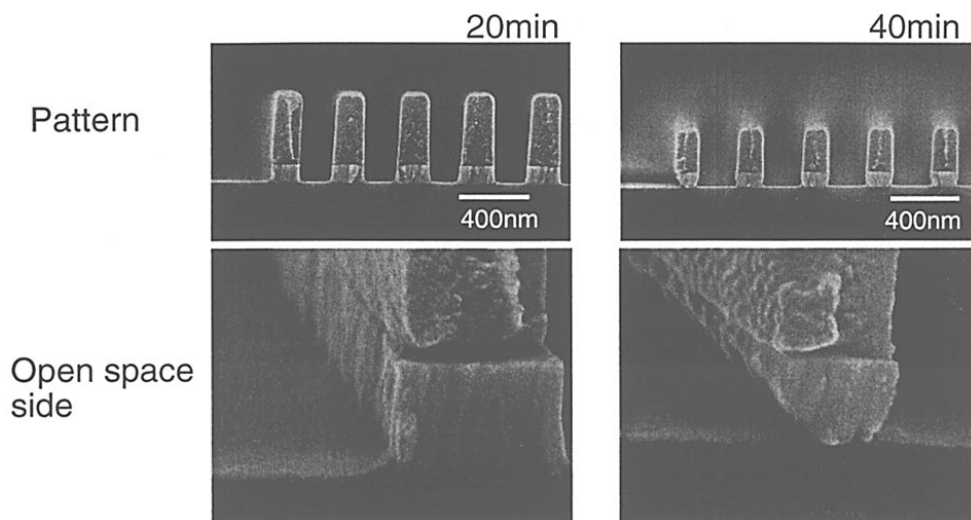


Figure 11. SEM photographs of Cr L&S patterns etched in $\text{Cl}_2/0.5\% \text{O}_2$ downstream plasmas for 20 min (left) and 40 min (right) at $V_{p-p} = 35 \text{ V}$.

tance, and serious swelling or shrinking occur when they were exposed to high flux electrons in the plasma. In this study we utilized the downstream Cl_2 plasmas where electrons are depleted through negative ion production. Consequently necessary radical and positive ions for etch reactions and sidewall protection can be introduced onto the substrate with the reduced electron flux 0.1-0.01 times lower than those in typical process plasmas and vertical Cr etching of $0.2 \mu\text{m}$ wide L&S patterns has been successfully attained using EB resists.

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